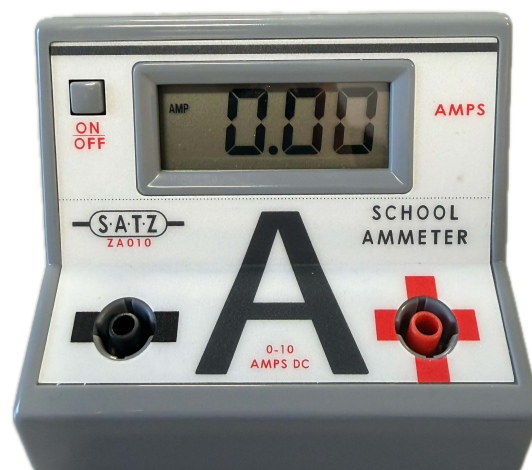




WJEC Physics Unit 5

2026 1A

Assessment Guide



Plan

- Rearrange to find linear relationship

$$R = k V^n$$

$$\ln R = n \ln V + \ln k$$

$$y = m x + c$$

- State what to plot

$\ln R$ against $\ln V$

- Determine value

slope = n, y intercept = $\ln k$

- Describe measurement taken

Adjust rheostat and record V and I

Quote: $R = \frac{V}{I}$

5 set of reading should be taken

The bulb might be hot so switch off the power supply after each reading

- Range

$$V_{\min} = 2.00V \quad I = 0.8A \quad R = 2.5\Omega$$

$$V_{\max} = 10.00V \quad I = 20A \quad R = 0.5\Omega$$

- List out values taken

Use range from 0 to 12V

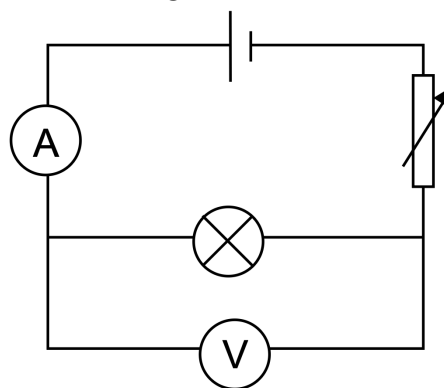
Approximately 2.00, 4.00, 6.00, 8.00, 10.00 V taken (5 set)

- Repeat readings are not needed for this task.

Sample Result

Voltage / V	Current / A
± 0.01	± 0.01
2.04	0.81
4.25	1.15
6.10	1.31
8.30	1.65
10.34	1.86

Circuit Diagram



Risk Assessment

Hazard	Risk	Control Measure
Hot lamp (can burn)	Touching the lamp and burning skin	Leave the lamp to cool before moving it / avoid touching

Process Result

Voltage (V) / V	Current (I) / A	R / Ω	ln V	ln R
± 0.01	± 0.01			
2.04	0.81	2.52	0.713	0.924
4.25	1.15	3.70	1.447	1.308
6.10	1.31	4.66	1.808	1.539
8.30	1.65	5.03	2.116	1.615
10.34	1.86	5.56	2.336	1.716

Graph

- ln R against ln V plotted
- Axis label shown (No unit shown)
- Suitable scales (points occupy $> \frac{1}{2}$ graph)
- All points plotted correctly
- Line of best fit correct

Questions:

1. Use your graph to determine suitable values for k and n. You are not required to calculate any uncertainties in these values. [4]
2. Write an equation for the relationship between resistance and potential difference for your filament lamp using your values for n and k. [1]
3. A student suggests that the same equation $R = k V^n$ will apply for the combination of two of these lamps connected in parallel. Discuss whether this students' suggestion is correct. [4]
 - Current for the circuit doubles when connected in parallel
 - Hence k has a different value
 - k will be halved
 - n stays the same

(c) Draw a suitable graph to determine n and k . Error bars are not required.

[4]

Examiner
only

